SANKAJITH D. JINASENA

Data Science Undergraduate

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PROFILE

Data Science undergraduate at Sabaragamuwa University of Sri Lanka with a strong track record of engineering end-to-end machine learning pipelines from raw data ingestion to production deployment. Combines solid statistical foundations with practical full-stack execution in Python, fine-tuned transformer architectures, and interactive dashboards. Delivers immediate engineering velocity to data teams by unblocking technical bottlenecks, accelerating model prototyping, and translating complex algorithmic outputs into reliable business insights.

EDUCATION

BSc (Hons) in Data Science - *Sabaragamuwa University of Sri Lanka* 2024 - Present

CGPA: 3.76

Coursework spanning machine learning, artificial intelligence, data analytics, and software engineering, applied through academic and independent projects.

GCE Advanced Level Examination - Physical Science Stream - *Dharmaraja College, Kandy* 2021 - 2022

Z-score: 1.3509

SKILLS

Technical

Programming Languages	Python, Java, PHP, SQL
Machine Learning & AI	Scikit-learn, TensorFlow, PyTorch
Data Analysis & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Power BI, Excel
Databases	MySQL, MongoDB
Cloud & Data Services	AWS S3, Lambda, Glue, Athena, CloudWatch
Web Development	HTML, CSS, JavaScript, React
Tools	Git, Docker, Microsoft Office, Filmora, Canva

Soft Skills & Languages

Soft Skills	Accountability & Ownership, Analytical Problem-Solving, Rapid Adaptability, Cross-Functional Collaboration, Technical Communication
Spoken Languages	English (Intermediate), Sinhala (Native/Bilingual)

EXPERIENCE

Digital Service Assistant - *Commercial Bank of Ceylon PLC* Oct 2023 - Apr 2024

Promoted and supported adoption of Commercial Bank's digital services (ComBank Digital, Q+, and Flash apps), educating clients on digital banking, providing technical and customer support, tracking usage metrics, and marketing digital platforms to expand customer engagement.

IT Consultant - *Dharmavijaya Computer Training School* Jan 2023 - Feb 2024

Delivered training and support in Microsoft Office, basic programming (Python, HTML, CSS), video editing, and computer hardware, while strengthening technical and managerial skills through supporting students and day-to-day operations.

PROJECTS

CareerCompass AI - Autonomous Career Coaching System Aug 2026

Tech: *Python, FastAPI, LangGraph, LangChain, Groq LPU (Llama 3.3), ChromaDB, React, Tailwind CSS, Pydantic, ReportLab*

- Engineered a 5-agent autonomous coaching system using a compiled LangGraph state machine to deliver end-to-end skill-gap, interview, and job matching reports.
- Implemented self-healing and adaptive routing loops via Pydantic validation to auto-correct JSON schemas and dynamically revise learning paths based on mock interview performance.
- Integrated dynamic fallback retrieval, combining ChromaDB vector search with real-time web scraping to synthesize skills taxonomies and fetch live job/course listings.

Detecting AI-Generated Product Reviews July 2026 - Sep 2026

Tech: *Python, NLP, TF-IDF, DistilBERT, React, FastAPI, Scikit-learn*

- Conducted a comparative NLP study across stylometrics, TF-IDF, and deep learning, fine-tuning a DistilBERT transformer that achieved 98.05% accuracy and a 0.981 F1-score in detecting AI-generated Amazon reviews.
- Engineered a 90.89% accurate feature-fusion model combining lexical and syntactic signals, deploying the full inference pipeline through a FastAPI backend and an interactive React dashboard.

Telecom Customer Churn Analytics & Predictive Retention System Oct 2026

Tech: *Python, XGBoost, Scikit-learn, SMOTE, AWS (S3, Lambda, Glue, Athena, CloudWatch), FastAPI, React*

- Engineered a predictive churn pipeline using SMOTE and Focal Loss XGBoost to handle severe class imbalance, achieving 80.21% Recall and 0.8468 ROC-AUC across 7,043 customer accounts.
- Architected an event-driven AWS analytics solution integrating S3 and Lambda for automated validation, alongside Glue and Amazon Athena serverless SQL to uncover critical churn segments accounting for $139k/month in lost revenue.

Sri Lanka Residential Electricity Affordability Analysis Aug 2026

Tech: *Power BI, DAX, Star Schema, Data Modeling*

- Built a 3-page interactive Power BI dashboard analyzing electricity consumption and affordability across 4,063 Sri Lankan households (2022-2025), uncovering income-based energy burden, disconnection risk, billing-awareness gaps, and smart-meter adoption patterns.

Employee Attrition Prediction using CRISP-DM Jun 2026

Tech: *Python, Pandas, Scikit-learn, CRISP-DM, Classification (Logistic Regression, Random Forest, SVM)*

- Executed the 6-stage CRISP-DM methodology on 1,470 employee records, engineering custom career-stage features and scaling features to address class imbalance.
- Benchmarked 5 classification algorithms, selecting Logistic Regression as the optimal operational model with 0.814 ROC-AUC and 0.610 Recall to prioritize high-risk retention candidates and pinpoint overtime and compensation as primary turnover drivers.

WellTrackeR – AI-Powered Personal Health Monitoring & Diagnostic System Sep 2025 - Nov 2025

Tech: *Python, Flask, Scikit-learn, PHP, MySQL, Chart.js, Dialogflow, JavaScript*

- Engineered a 3-tier, role-based clinical monitoring platform supporting Patients, Doctors, and Admins across a normalized 10-table MySQL relational schema with Chart.js vitals tracking and automated PDF health reporting.
- Integrated multi-condition ML risk screening by serving Scikit-learn disease prediction models (diabetes, lung cancer, brain tumor) via a Flask REST microservice paired with a conversational Dialogflow triage assistant.

Microsoft Stock Probability Analysis Sep 2025 - Nov 2025

Tech: *Python (Pandas, NumPy, Matplotlib, Seaborn), SciPy, Statsmodels, Monte Carlo Simulation, Markov Chains*

- Modeled 39 years of daily historical stock returns (1986-2025) using parametric distributions (Normal, Log-Normal, Student's t), fitting heavy-tailed characteristics and evaluating empirical win-loss probabilities.
- Implemented Markov chain transition matrices and Monte Carlo simulations to quantify trend momentum, conditional downside probability, and portfolio risk metrics including Value at Risk (VaR), Conditional VaR (CVaR), and Sharpe Ratio.

AWARDS & ACHIEVEMENTS

Codemania V6.0 Datathon - Top 15 Finalist *(Team Deadlock)*

- Selected from a competitive pool of inter-university participants as a Top 15 Finalist, applying data-driven approaches to solve complex problems in a fast-paced environment.

CryptX 2.0 - Top 15 Finalist *(Team Predictra)*

- Finalist at the CryptX 2.0 hackathon organized by the ICT Society, Faculty of Technology, University of Sri Jayewardenepura.

BitCode 6.0 - Top 10 Finalist *(Team Deadlock)*

- Finalist at the BitCode 6.0 hackathon organized by the Business Information Technology Students' Association, Rajarata University.

J'PuraXtreme 3.0 - Finalist *(Team Predictra)*

- Qualified for the final round, demonstrating strong problem-solving, coding, logical thinking, and teamwork under time pressure.

CIPHER 2.0 - Top 10 Finalist *(Team Predictra)*

- Finalist among top 10 teams at CIPHER 2.0, with the final pitch held at the IIT Auditorium.

CodeAréna'26 - Finalist *(Team Predictra)*

- Advanced to the Grand Finale following a 36-hour developing, debugging, and pitching a complete solution under tight time constraints at Dialog Axiata PLC Headquarters.

CodeRally 7.0 - Intermediate Tier Grand Finalist *(Team Predictra)*

- Qualified for the Intermediate Tier Grand Finale and a 24-hour hackathon challenge, demonstrating strong technical problem-solving, adaptability, and teamwork under pressure.

CERTIFICATES

- Data Scientist Associate Certificate - DataCamp (*Expires: 28 October 2028*)
- Python Programming - CODL, University of Moratuwa
- Google AI Essentials - Google (Coursera)
- ML for Beginners - Sololearn
- Introduction to Docker - DataCamp
- Web Design for Beginners - CODL, University of Moratuwa
- Fundamentals of Digital Marketing - Google
- Foundations of Project Management - CODL, University of Moratuwa

VOLUNTEERING

STEMUP Volunteer - ***STEMUP Educational Foundation*** 2025 - Present

- Supported STEM education activities and assisted in organizing learning sessions for students, contributing to community-based technology and education initiatives.

Volunteer - ***Rotaract Club of SUSL*** 2025 - Present

- Contributing mainly on the technical side, supporting the club 's IT needs and activities.

Volunteer - IEEE Student Branch, Sabaragamuwa University of Sri Lanka 2026 - Present

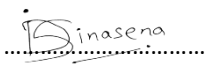
- Support the technical coordination of IEEE activities and events, including IT-related tasks, digital platforms, and other technical requirements.

REFERENCES

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I hereby assert that the information provided above is true and accurate to the best of my knowledge.



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